

11-dars

Date,

No.

Tavcorloshi Project 2 / Feature Engineering (Techniques) / Class (60P)
Institution facilities: Fold-cross validation / Standard & Creativity

Model Building Project

Project 2 (Dataset groupada)

- Dataset (car 2018 csv)
- Eng yatshi accuracy / F2 / mac
- Eng tashkili: Eng tshiq Eng tshumari:

--- Feature Engineering ---
(Inputlar bilan ishlash)

Modelni Rivojlantirish (Model Building ishlar)

- Yangi featurelar yaratish, ularni ber tushun ivetishi
- Yangi transform qilish va muhim featurelarni tanlash bosqichi
- Model Performansini yaxshilash ^{pattern} generalizatsionini oshiradi

Feature Engineering (Techniques)

- Handling missing values
- Encoding
- Scaling (Normalization) → Transforming
- Feature Creation
- Feature Selection

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- Feature Creation:

~ Dadooni Pau data yetarli bo'lmaganda

~ Agar kuz va foydali data bo'lsa model yaratma-vetlikini (patterns)ni yaxshi tushunadi.

- Nimalar yarat

• Domain knowledge (featurelarni bilish)

• Creativity (yaratuvchilik)

• Statistical insight (Statistical ma'lumotni fohish)

Feature Engineering Techniques

• Hima uchun mahiri

- Modelni predictive qobiliyatini oshiradi

• Teleson PATRI, rangga kora uni modelni yaratish

- Model o'rganishni osonlashtirish

• Kichik featurelarni birlashtirish (featurelarni + o'z-aro + family)

- Hidden Relationships:

• Butun ma'lumotdagi g'alloar soni o'rtacha xar bor

o'rtacha o'rtacha g'alloar soni faqatgina (g'alloar total goals / matches)

• Danday bog'lanishi:

- Matematik yul o'rali (+, -, /, Sum, mean)

- Domain knowledge

• House age = Current year - Year built

- Time-based

• 2025.09.02 = 2025/09/02

o'rtacha birinchi-birinchi ushbu usulda fohish.

• Formula:

Feature Engineering Analogy.

#Xas bir oyun uclun ortoluk goller soni:

```
df['ggg's'] = df['goals's'] / df['games's']
df.info()
```

#Sabaki oymulini stbkilligun vortsh

```
df['ggg's'] = df['yellow-cards's'] / df['games's']
df.info()
```

#Tutizon boydun yugui feature

```
df['XGDIFF's'] = df['xG's'] - df['goals's']
df.info()
```

--- Class ---

• Blueprint / Template == Plan

• Object == Real Mouse

• Function 1 + function 2 + function 3 = Class

- Class Analogy:

#Funksiya

```
def a-b-add(a, b):
```

return a+b

↓
a-b-add(1, 2)

'''3

a-b-add(1, 11)

'''

class Xisobla:

```
def __init__(self, a, b):
```

self.a = a

self.b = b

```
def add(self):
```

return self.a + self.b

notifa = Xisobla(1, 2)

notifa.add()

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class Xisobla:

def __init__(self, a, b):

self.a = a

self.b = b

def add(self):

return self.a + self.b

def isop(self):

return self.a * self.b

def orta(self):

return (self.a + self.b) / 2

x = Xisobla(1, 2)

x.add()

x.isop()

x.orta()